Fault-free Paths in Hypercubes with More Faulty Nodes

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Abstract

Faults in a network may take various forms such as hardware/software errors, node/link faults, etc. In this paper, node-faults are addressed. Let F be a faulty set of $f \leq 2n-6$ conditional node-faults in an injured n-cube Q_n such that every node of Q_n still has at least two fault-free neighbors. Then we show that $Q_n - F$ contains a path of length at least $2^n - 2f - 1$ (respectively, $2^n - 2f - 2$) between any two nodes of odd (respectively, even) distance. Since an n-cube is a bipartite graph, such kind of the fault-free path turns out to be the longest one in the case when all faulty nodes belong to the same partite set.

1 Introduction

In many parallel computer systems, processors are connected on the basis of interconnection networks such as hypercubes, star graphs, meshes, bubble-sort networks, etc. For the sake of simplicity, a network topology can be modeled by a graph, in which nodes correspond to processors and links correspond to communication channels. Throughout this paper, we concentrate on loopless undirected graphs. For the graph definitions and notations we follow the ones given by Bondy and Murty [1]. A graph G consists of a set $V(G)$ and a subset $E(G)$ of $\{(u,v) \mid (u,v)$ is an unordered pair of $V(G)\}$. The set $V(G)$ is called the *node set* of G and $E(G)$ is called the *link set* of G. Two nodes u and v of G are adjacent if $(u,v) \in E(G)$. For a node u of G, its *neighborhood* $N_G(u)$ is defined by $N_G(u) = \{v \in V(G) \mid (u,v) \in E(G)\}$. A graph H is a *subgraph* of G if $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$. A graph G is *bipartite* if its node set can be partitioned into two disjoint partite sets $V_0(G)$ and $V_1(G)$ such that every link will join a node of $V_0(G)$ and a node of $V_1(G)$.

A *path* P of length k from node x to node y in a graph G is a sequence of distinct nodes $\langle v_1, v_2, \ldots, v_{k+1}\rangle$ such that $x = v_1$, $y = v_{k+1}$, and $(v_i, v_{i+1}) \in E(G)$ for every $1 \leq i \leq k$ if $k \geq 1$. For convenience, we write P as $\langle v_1, \ldots, v_i, Q, v_j, \ldots, v_{k+1}\rangle$ where $Q = \langle v_i, v_{i+1}, \ldots, v_j\rangle$. Note that we allow Q to be a path of length zero. The i-th node of P is denoted by $P(i)$; i.e., $P(i) = v_i$. Moreover, we use P^{-1} to denote the path $\langle v_{k+1}, v_k, \ldots, v_2, v_1\rangle$. For two nodes u and v of G, the *distance* between u and v, denoted by $d_G(u,v)$, is the minimum length among paths of G joining u and v. A *cycle* is a path with at least three nodes such that the last node is adjacent to the first one. For clarity, a cycle of length k is represented by $\langle v_1, v_2, \ldots, v_k, v_1\rangle$. A path of a graph G is a *hamiltonian path* if it spans G. Similarly, a cycle of a graph G is a *hamiltonian cycle* if it spans G. A bipartite graph is *hamiltonian laceable* [12] if there is a hamiltonian path between any two nodes which are in different partite sets. Moreover, a hamiltonian laceable graph G is *hyper hamiltonian laceable* [9] if for any node $v \in V_i(G)$, $i \in \{0,1\}$, there is a hamiltonian path of $G - \{v\}$ between any two nodes of $V_{1-i}(G)$. Later Hsieh et al. [5] introduced the *strongly hamiltonian laceability.* A hamiltonian laceable graph G is strongly hamiltonian laceable if there is a path of length $|V(G)| - 2$ between any two nodes in the same partite set.

Hypercube is one of the most popular interconnection networks discovered for parallel computation. The *n-dimensional hypercube* (or n-

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cube for short) is defined as follows. Let $u = b_{n-1} \dots b_i \dots b_0$ be an n-bit binary string. For any i, $0 \leq i \leq n-1$, we use $(u)^i$ to denote the binary string $b_{n-1} \dots \bar{b}_i \dots b_0$. Moreover, we use $(u)_i$ to denote the bit b_i of u. The *Hamming weight* of u, denoted by $w_H(u)$, is $|\{j \mid (u)_j = 1, 0 \leq j \leq n-1\}|$. The n-cube Q_n consists of all n-bit binary strings representing its nodes. Two nodes u and v are adjacent if and only if $v = (u)^i$ for some i and we call the link $(u, (u)^i)$ an i-dimensional link. We define $dim((u,v)) = i$ if and only if $v = (u)^i$. The *Hamming distance* between u and v, denoted by $h(u,v)$, is defined to be $|\{j \mid (u)_j \neq (v)_j, 0 \leq j \leq n-1\}|$. Hence, two nodes u and v are adjacent if and only if $h(u,v) = 1$. It is well known that Q_n is a bipartite graph with partite sets $V_0(Q_n) = \{u \in V(Q_n) \mid w_H(u) \text{ is even}\}$ and $V_1(Q_n) = \{u \in V(Q_n) \mid w_H(u) \text{ is odd}\}$. It is also known that Q_n is node-transitive and link-transitive.

Because nodes and/or links in a network may fail accidentally, it is demanded to consider the fault-tolerance of a network. Hence, the issue of faulty hypercubes has been widely addressed in the literature [2,3,6,7,10,11,13–15]. For example, Tseng [15] showed that a faulty n-cube, containing $f_e \leq n-4$ faulty links and $f_v \leq n-1$ faulty nodes with $f_e + f_v \leq n-1$, has a fault-free cycle of length at least $2^n - 2f_v$. Fu [2] and Hsieh [6] further studied the fault-tolerant cycle embedding in hypercubes. On the other hand, Fu [3] investigated the longest paths in an n-cube with up to $n-2$ faulty nodes. Recently, Ma et al. [11] proved that a faulty n-cube, containing f_e faulty links and f_v faulty nodes with $f_e + f_v \leq n-2$, has a fault-free path of length l between any two fault-free nodes for every l satisfying $d_{Q_n}(u,v) + 2 \leq l \leq 2^n - 2f_v - 1$ and $l - d_{Q_n}(u,v) \equiv 0 \pmod 2$. However, one should notice that each component in a network may have independent reliability. That is, if components of a network fail independently, the probability that all failures would be close to each other becomes low. Due to this motivation, Harary [4] first introduced the idea of *conditional connectivity.* Later, Latifi et al. [8] defined the *conditional node-faults* which require each node of a network to have at least g fault-free neighbors, $g \geq 1$. In this paper, we focus on $g = 2$ and we say a network is *conditionally faulty* if its every node has at least two fault-free neighbors. Under this premise, we improve Fu's result [3] by showing that an n-cube with $f \leq 2n-6$ conditional node-faults contains a fault-free path of length at least $2^n - 2f - 1$ (respectively, $2^n - 2f - 2$) between any two fault-free nodes of odd (respectively, even) distance. Since an n-cube is a bipartite graph, such kind of the fault-free path turns out to be the longest one in the case when all faulty nodes belong to the same partite set.

The condition of having at least two fault-free neighbors for every node is statistically reasonable. Suppose, with a so-called random fault model, the probability of node failure is identical and nodes fail independently. Let $Pr(n)$ denote the probability that every node of an n-cube containing $2n-6$ faulty nodes is adjacent to at least two fault-free neighbors. Clearly, $Pr(3) = Pr(4) = 1$ and $Pr(5) = 1 - \frac{2^5 \times \binom{5}{4}}{\binom{2^5}{4}} = \frac{895}{899}$, where $2^5 \times \binom{5}{4}$ is the number of node-fault distributions that some node has four faulty neighbors. When $n \geq 6$, the number of node-fault distributions that some node has n or $n-1$ faulty neighbors is $2^n \times \binom{2^n-n}{n-6} + 2^n \times \binom{n}{n-1}\binom{2^n-n}{n-5}$. Hence, we have $Pr(n) = 1 - \frac{2^n \times \binom{2^n-n}{n-6} + 2^n \times \binom{n}{n-1}\binom{2^n-n}{n-5}}{\binom{2^n}{2n-6}}$. It can be verified that $Pr(n)$ approaches to 1 as n increases.

The rest of this paper is organized as follows. In Section 2, the partition of an n-cube with conditional node-faults is proposed. In Section 3, we show that an n-cube with $f \leq 2n-6$ conditional node-faults contains a fault-free path of length at least $2^n - 2f - 1$ (respectively, $2^n - 2f - 2$) between any two fault-free nodes of odd (respectively, even) distance. Finally, the conclusion is given in Section 4.

2 Partition of an n-cube with conditional node-faults

For $0 \leq i \leq n-1$ and $j \in \{0,1\}$, let $Q_n^{i,j}$ be the subgraph of Q_n induced by $\{u \in V(Q_n) \mid (u)_i = j\}$. Obviously, $Q_n^{i,j}$ is isomorphic to Q_{n-1}. Then an *i-partition* of Q_n divides Q_n along dimension i into $\{Q_n^{i,0}, Q_n^{i,1}\}$. In order to clearly indicate the faulty elements in a network G, we use $F(G)$ to denote the set of all faulty elements in G. Let u be a node of G. For convenience, we use $N_G^F(u)$ to denote the set of all faulty neighbors of u; i.e., $N_G^F(u) = N_G(u) \cap F(G)$.

Suppose Q_n ($n \geq 5$) contains $f \leq 2n-6$ faulty nodes such that its every node has at least two fault-free neighbors. Moreover, suppose that u, v, and w are three nodes of this faulty n-cube and each of them has only two fault-free neighbors. In what follows, we discuss how the faulty nodes will be located conditionally. It is noted that any two nodes of an n-cube have at most two common neighbors. If $|N^F_{Q_n}(v) \cap N^F_{Q_n}(w)| \leq 1$, then $f \geq |N^F_{Q_n}(v) \cup N^F_{Q_n}(w)| = |N^F_{Q_n}(v)| + |N^F_{Q_n}(w)| - |N^F_{Q_n}(v) \cap N^F_{Q_n}(w)| \geq 2(n-2) - 1 = 2n-5$ contradicts the requirement that $f \leq 2n-6$. Therefore, $|N^F_{Q_n}(v) \cap N^F_{Q_n}(w)| = 2$ must be satisfied. Similarly, we have $|N^F_{Q_n}(u) \cap N^F_{Q_n}(v)| = 2$ and $|N^F_{Q_n}(u) \cap N^F_{Q_n}(w)| = 2$. Then we consider $|N^F_{Q_n}(u) \cap N^F_{Q_n}(v) \cap N^F_{Q_n}(w)| \in \{0, 1\}$. With respect to the topological structure of an n-cube, it is not difficult to see that $|N^F_{Q_n}(u) \cap N^F_{Q_n}(v) \cap N^F_{Q_n}(w)| = 0$ is impossible. Accordingly, we have $2n-6 \geq f \geq |N^F_{Q_n}(u) \cup N^F_{Q_n}(v) \cup N^F_{Q_n}(w)| = 3(n-2) - 3 \times 2 + 1 = 3n - 11$; that is, $n = 5$. Therefore, we have the following lemma.

Lemma 1. *Let an n-cube Q_n ($n \geq 5$) be conditionally faulty and contain $f \leq 2n-6$ faulty nodes. Suppose that u, v and w are three nodes of Q_n and each of them has exactly two fault-free neighbors. Then n must equal 5 and there exists a dimension j of $\{0, 1, \dots, n-1\}$ so that both $Q^{j,0}_n$ and $Q^{j,1}_n$ are conditionally faulty. Moreover, both $Q^{j,0}_n$ and $Q^{j,1}_n$ contain at most $2n-8$ faulty nodes.*

Proof. Since $|N^F_{Q_n}(u) \cup N^F_{Q_n}(v) \cup N^F_{Q_n}(w)| \leq f \leq 2n-6$, n must equal 5. By brute force, the faulty nodes will be distributed as that shown in Figure 1(a). Let k_1 and k_2 be two integers of $\{0, 1, 2, 3, 4\}$ such that both $(u)^{k_1}$ and $(u)^{k_2}$ are fault-free and let $\{k_3, k_4, k_5\} = \{0, 1, 2, 3, 4\} - \{k_1, k_2\}$. Then Q_5 can be partitioned along any dimension $j \in \{k_3, k_4, k_5\}$ so that both $Q^{j,0}_5$ and $Q^{j,1}_5$ are conditionally faulty. Obviously, $Q^{j,0}_5$ and $Q^{j,1}_5$ contain two faulty nodes. □

Lemma 2. *Let an n-cube Q_n ($n \geq 5$) be conditionally faulty and contain $f \leq 2n-6$ faulty nodes. Suppose that u and v are two nodes of Q_n and only u and v have exactly two fault-free neighbors. Then there exists a dimension k of $\{0, 1, \dots, n-1\}$ so that both $Q^{k,0}_n$ and $Q^{k,1}_n$ are conditionally faulty. Moreover, both $Q^{k,0}_n$ and $Q^{k,1}_n$ contain at most $2n-8$ faulty nodes.*

Proof. Since both u and v have $n-2$ faulty neighbors, they must have at least two common faulty neighbors. Since any two nodes of Q_n have at most two common neighbors, u and v have exactly two faulty neighbors in common. Accordingly, let i and j be two integers of $\{0, 1, \dots, n-1\}$ such that $(u)^i$ and $(u)^j$ are just the common faulty neighbors of u and v. Obviously, we have $(u)^i = (v)^j$ and $(u)^j = (v)^i$. Then we can partition Q_n along dimension $k \in \{i, j\}$ so that both $(n-1)$-dimensional subcubes are conditionally faulty. Obviously, both $Q^{k,0}_n$ and $Q^{k,1}_n$ contain $n-3$ faulty nodes and we have $n-3 \leq 2n-8$ for $n \geq 5$. □

Lemma 3. *Suppose that an n-cube Q_n ($n \geq 5$) is conditionally faulty and contains $f \leq 2n-6$ faulty nodes. Let z be the unique node which has only two fault-free neighbors. Then there exists a dimension j of $\{0, 1, \dots, n-1\}$ so that both $Q^{j,0}_n$ and $Q^{j,1}_n$ are conditionally faulty. Moreover, both $Q^{j,0}_n$ and $Q^{j,1}_n$ will contain at most $2n-8$ faulty nodes except for the cases depicted in Figure 1(b,c,d).*

Proof. Let k_1 and k_2 be two integers such that $(z)^{k_1}$ and $(z)^{k_2}$ are fault-free. Moreover, let $\{k_3, k_4, \dots, k_n\} = \{0, 1, \dots, n-1\} - \{k_1, k_2\}$. For convenience, let F denote the faulty set of Q_n. Obviously, every dimension i of $\{k_3, k_4, \dots, k_n\}$ can be used to partition Q_n so that both $Q^{i,0}_n$ and $Q^{i,1}_n$ are conditionally faulty. Suppose $f \leq 2n-7$. Obviously, there exists a dimension i of $\{k_3, k_4, \dots, k_n\}$ so that both $Q^{i,0}_n$ and $Q^{i,1}_n$ contain no more than $2n-8$ faulty nodes. Thus, we assume $f = 2n-6$ in the following discussion. Suppose that there does not exist any dimension i of $\{k_3, k_4, \dots, k_n\}$ so that both $Q^{i,0}_n$ and $Q^{i,1}_n$ contain $2n-8$ or less faulty nodes. Then node z and all of $F - \{(z)^{k_3}, (z)^{k_4}, \dots, (z)^{k_n}\}$ are located on $Q^{i,(z)_i}_n$ for each $i \in \{k_3, k_4, \dots, k_n\}$. Hence, every node x of $F - \{(z)^{k_3}, (z)^{k_4}, \dots, (z)^{k_n}\}$ satisfies $(x)_i = (z)_i$ for each $i \in \{k_3, k_4, \dots, k_n\}$. Thus, we have $F - \{(z)^{k_3}, (z)^{k_4}, \dots, (z)^{k_n}\} \subseteq \{z, ((z)^{k_1})^{k_2}\}$. Since $|F - \{(z)^{k_3}, (z)^{k_4}, \dots, (z)^{k_n}\}| = f - (n-2) = (2n-6) - (n-2) \leq 2 = |\{z, ((z)^{k_1})^{k_2}\}|$, we have $n \leq 6$. That is, if $n \geq 7$, there exists a dimension j of $\{k_3, k_4, \dots, k_n\}$ so that both $Q^{j,0}_n$ and $Q^{j,1}_n$ contain $2n-8$ or less faulty nodes. For $n = 5$, the cases are depicted in Figure 1(b,c). Obviously, every dimension i of $\{k_3, k_4, k_5\}$ results in $||F(Q^{i,0}_5)| - |F(Q^{i,1}_5)|| = 2$. For $n = 6$, the case is depicted in Figure 1(d). Similarly, every dimension i of $\{k_3, k_4, k_5, k_6\}$ results in $||F(Q^{i,0}_6)| - |F(Q^{i,1}_6)|| = 4$. □

Lemma 4. *Suppose that an n-cube Q_n ($n \geq 6$) contains $f \leq 2n-6$ faulty nodes such that its every node has at least three fault-free neighbors. Then there exists a dimension j of $\{0, 1, \dots, n-1\}$ so that not only are $Q^{j,0}_n$ and $Q^{j,1}_n$ conditionally faulty but also they contain at most $2n-8$ faulty nodes.*

Proof. Obviously, every $(n-1)$-dimensional subcube of Q_n is conditionally faulty. Moreover, we only need to consider $f = 2n-6$. Suppose, by contradiction, that the result is not true. Since Q_n is

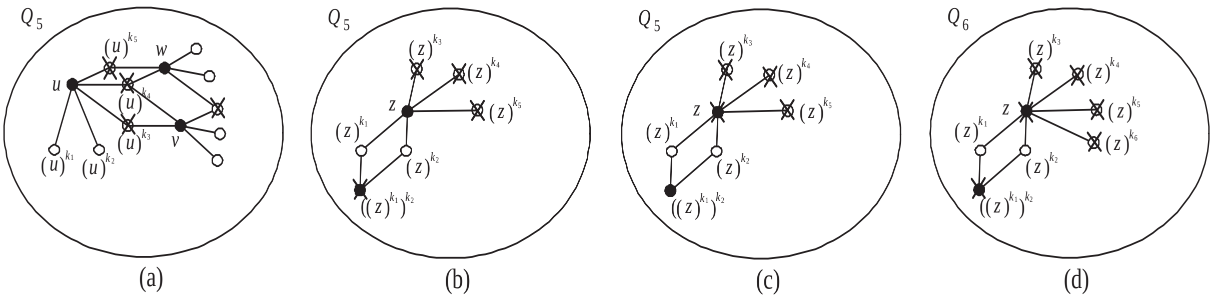

Figure 1: Every faulty node is marked by an "X" symbol. (a) The distribution of conditional node-faults in Q_5 so that each of three nodes u, v and w has only two fault-free neighbors; (b) The distribution of conditional node-faults in Q_5 so that every dimension i of $\{k_3, k_4, k_5\}$ results in $||F(Q_5^{i,0})| - |F(Q_5^{i,1})|| = 2$ while z is fault-free; (c) The distribution of conditional node-faults in Q_5 so that every dimension i of $\{k_3, k_4, k_5\}$ results in $||F(Q_5^{i,0})| - |F(Q_5^{i,1})|| = 2$ while z is faulty; (d) The distribution of conditional node-faults in Q_6 so that every dimension i of $\{k_3, k_4, k_5, k_6\}$ results in $||F(Q_6^{i,0})| - |F(Q_6^{i,1})|| = 4$.

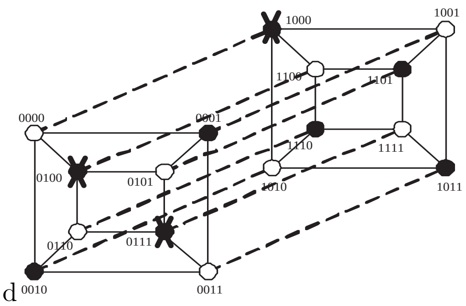

Figure 2: A 4-cube, with 3 conditional node-faults, cannot be partitioned along any dimension in such a way that both 3-dimensional subcubes are conditionally faulty. Every faulty node is marked by an "X" symbol.

node-transitive, we can assume $\mathbf{e} = 0^n$ is a faulty node. Also, we assume that all faulty nodes have the same bits in exactly q dimensions. Thus, the remaining $n-q$ dimensions, say $\{0, 1, \ldots, n-q-1\}$, can be used to partition Q_n so that exactly one subcube contains only one faulty node. Without loss of generality, we further assume there are exactly p dimensions of $\{0, 1, \ldots, n-q-1\}$, say $\{0, \ldots, p-1\}$, such that $\mathbf{e}$ is the unique faulty node located in $Q_n^{i,0}$ for all $0 \le i \le p-1$. By the node-transitivity, we have $1 \le p \le n-q$. Then it is not hard to see that the node $0^{n-p}1^p$ happens to have at least $n-2$ faulty neighbors, contradicting the initial requirement. Hence, the theorem is proved. $\square$

Figure 2 is an example of the n-cube with $2n-5$ conditional node-faults and it cannot be partitioned along any dimension in such a way that both $(n-1)$-dimensional subcubes are conditionally faulty. For this reason, our attention is confined to $2n-6$ faulty nodes.

3 Longest paths in hypercubes with conditional node-faults

The following theorems concern the hamiltonian paths on hypercubes with faulty links.

Theorem 1. *[13] Let $n \ge 3$. Suppose that $F \subseteq E(Q_n)$ is a set of at most $n-2$ faulty links. Then $Q_n - F$ is hamiltonian laceable and strongly hamiltonian laceable.*

Theorem 2. *[13] Let $n \ge 3$. Suppose that $F \subseteq E(Q_n)$ is a set of at most $n-3$ faulty links. Then $Q_n - F$ is hyper hamiltonian laceable.*

As Fu [3] showed, an n-cube with $f \le n-2$ faulty links has a fault-free path of length at least $2^n - 2f - 1$ (respectively, $2^n - 2f - 2$) between two arbitrary nodes of odd (respectively, even) distance.

Theorem 3. *[3] Suppose $n \ge 3$. Let u and v be two arbitrary fault-free nodes in an n-cube with $f \le n-2$ faulty nodes. If $h(u, v)$ is odd, then there exists a fault-free path of length at least $2^n - 2f - 1$ joining u and v. Otherwise, there exists a fault-free path of length at least $2^n - 2f - 2$ between u and v.*

Now we show that an n-cube can tolerate more faulty nodes under the assumption of conditional node-faults.

Theorem 4. *Let F be a set of $f \le 2n-6$ conditional node-faults in Q_n ($n \ge 4$) such that every node of Q_n has at least two fault-free neighbors. Suppose that s and t are two arbitrary nodes of $Q_n - F$. Then $Q_n - F$ contains a path of length at least $2^n - 2f - 1$ (respectively, $2^n - 2f - 2$) between s and t if $h(s,t)$ is odd (respectively, even).*

Proof. By Theorem 3, this statement is true for $n = 4$. Thus, we only need to consider $n \ge 5$. With Lemmas 1$\sim$4, there exists a j-partition of Q_n, $n \ge 6$, so that both $Q_n^{j,0}$ and $Q_n^{j,1}$ are conditionally faulty. As a result, the proof can proceed by induction on n. The induction base, depending upon Q_5, can be checked by a computer program. As the inductive hypothesis, we assume that the theorem holds for Q_{n-1} when $n \ge 6$. For convenience, let f_0 (respectively, f_1) denote the number of faulty nodes in $Q_n^{j,0}$ (respectively, $Q_n^{j,1}$). Obviously, $f = f_0 + f_1$. Then we consider the following two cases.

Case I: Suppose that $h(s,t)$ is odd. Without loss of generality, we assume $s \in V_0(Q_n - F)$ and $t \in V_1(Q_n - F)$.

Subcase I.1: Suppose that both f_0 and f_1 are $2n-8$ or less. With symmetry, we assume that s is in $Q_n^{j,0}$.

Subcase I.1.1: Suppose that t is in $Q_n^{j,0}$. By the inductive hypothesis, $Q_n^{j,0} - F(Q_n^{j,0})$ contains a path H_0 of length at least $2^{n-1} - 2f_0 - 1$ between s and t. Let $A = \{(H_0(i), H_0(i+1)) \mid 1 \le i \le 2^{n-1} - 2f_0$ and i is odd$\}$ be a set of $\lceil \frac{2^{n-1}-2f_0-1}{2} \rceil$ links on H_0. Since $\lceil \frac{2^{n-1}-2f_0-1}{2} \rceil > 2n-6-f_0 \ge f_1$ for any $n \ge 6$, there exists a link (x,y) of A such that both $(x)^j$ and $(y)^j$ are fault-free. Hence, the path H_0 can be written as $\langle s, H_0', x, y, H_0'', t\rangle$. By the inductive hypothesis, $Q_n^{j,1} - F(Q_n^{j,1})$ contains a path H_1 of length at least $2^{n-1} - 2f_1 - 1$ between $(x)^j$ and $(y)^j$. Then $\langle s, H_0', x, (x)^j, H_1, (y)^j, y, H_0'', t\rangle$ is a fault-free path of length at least $2^n - 2f - 1$ between s and t. See Figure 3(a).

Subcase I.1.2: Suppose that t is in $Q_n^{j,1}$. Since $|V_1(Q_n^{j,0})| - f_0 = 2^{n-2} - f_0 > 2n-6-f_0 \ge f_1$ for $n \ge 6$, there exists a fault-free node b of $V_1(Q_n^{j,0})$ such that $(b)^j$ is also fault-free. By the inductive hypothesis, $Q_n^{j,0} - F(Q_n^{j,0})$ contains a path H_0 of length at least $2^{n-1} - 2f_0 - 1$ between s and b. Similarly, $Q_n^{j,1} - F(Q_n^{j,1})$ contains a path H_1 of length at least $2^{n-1} - 2f_1 - 1$ between $(b)^j$ and t. Then $\langle s, H_0, b, (b)^j, H_1, t\rangle$ is a fault-free path of length at least $2^n - 2f - 1$ between s and t. See Figure 3(b).

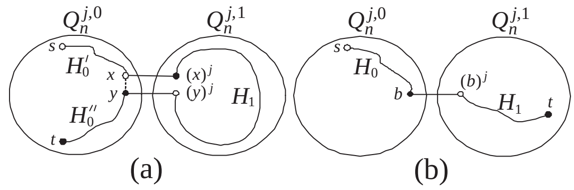

Figure 3: Illustration for Subcase I.1 of Theorem 4.

Subcase I.2: Suppose that either f_0 or f_1 is $2n-7$. With Lemmas 1$\sim$4, this case occurs for $n \in \{5,6\}$ only and the faulty nodes will be distributed as those illustrated in Figure 1(b,c,d). To be precise, let z be the only one node with $n-2$ faulty neighbors. Moreover, let k_1 and k_2 be two integers of $\{0,1,\ldots,n-1\}$ such that both $(z)^{k_1}$ and $(z)^{k_2}$ are fault-free. Accordingly, we can partition Q_n along dimension $j \in \{0,1,\ldots,n-1\} - \{k_1,k_2\}$. With symmetry, we assume $f_0 = F(Q_n^{j,0}) = 2n-7$ and thus we obtain $f_1 = f - f_0 = 1$. One should notice that $(z)^j$ is the unique faulty node in $Q_n^{j,1}$. Since $n \ge 6$, we only consider the case illustrated in Figure 1(d). As a consequence, it is straightforward to check this case by a computer program.

Case II: Suppose that $h(s,t)$ is even. Without loss of generality, we assume $s, t \in V_0(Q_n - F)$.

Subcase II.1: Suppose that both f_0 and f_1 are $2n-8$ or less. With symmetry, we assume that s is in $Q_n^{j,0}$.

Subcase II.1.1: Suppose that t is in $Q_n^{j,0}$. By the inductive hypothesis, $Q_n^{j,0} - F(Q_n^{j,0})$ contains a path H_0 of length at least $2^{n-1} - 2f_0 - 2$ between s and t. Let $A = \{(H_0(i), H_0(i+1)) \mid 1 \le i \le 2^{n-1} - 2f_0 - 1$ and i is odd$\}$ be a collection of $\lceil \frac{2^{n-1}-2f_0-2}{2} \rceil$ links on H_0. Since $\lceil \frac{2^{n-1}-2f_0-2}{2} \rceil > 2n-6-f_0 \ge f_1$ for any $n \ge 6$, there exists a link (x,y) of A such that both $(x)^j$ and $(y)^j$ are fault-free. Hence, the path H_0 can be written as $\langle s, H_0', x, y, H_0'', t\rangle$. By the inductive hypothesis, $Q_n^{j,1} - F(Q_n^{j,1})$ contains a path H_1 of length at least $2^{n-1} - 2f_1 - 1$ between $(x)^j$ and $(y)^j$. Then $\langle s, H_0', x, (x)^j, H_1, (y)^j, y, H_0'', t\rangle$ is a fault-free path of length at least $2^n - 2f - 2$ between s and t.

Subcase I.1.2: Suppose that t is in $Q_n^{j,1}$. Since $|V_1(Q_n^{j,0})| - f_0 = 2^{n-2} - f_0 > 2n-6-f_0 \ge f_1$ for $n \ge 6$, there exists a fault-free node b of $V_1(Q_n^{j,0})$ such that $(b)^j$ is also fault-free. By the inductive hypothesis, $Q_n^{j,0} - F(Q_n^{j,0})$ contains a path H_0 of length at least $2^{n-1} - 2f_0 - 1$ between s and b. Similarly, $Q_n^{j,1} - F(Q_n^{j,1})$ contains a path H_1 of length at least $2^{n-1} - 2f_1 - 2$ between $(b)^j$ and t. Then $\langle s, H_0, b, (b)^j, H_1, t\rangle$ is a fault-free path of length at least $2^n - 2f - 2$ between s and t.

Subcase II.2: Suppose that either f_0 or f_1 is $2n-7$. Similar to Subcase I.2, this case occurs for $n \in \{5, 6\}$ only and the faulty nodes will be distributed as those illustrated in Figure 1(b,c,d). Let z be the unique node which has $n-2$ faulty neighbors. Moreover, Let k_1 and k_2 be two integers of $\{0, 1, \ldots, n-1\}$ such that both $(z)^{k_1}$ and $(z)^{k_2}$ are fault-free. Accordingly, we partition Q_n along dimension $j \in \{0, 1, \ldots, n-1\} - \{k_1, k_2\}$. With symmetry, we assume $f_0 = 2n-7$ and $f_1 = 1$. Note that, $(z)^j$ is the unique faulty node in $Q_n^{j,1}$. Since $n \geq 6$, we only consider the case illustrated in Figure 1(d). As a consequence, it is straightforward to check this case by a computer program.

Therefore, the proof is completed. $\square$

4 Conclusions

As discussed above, the condition of requiring every node to have at least two fault-free neighbors is probabilistically reasonable. With Lemmas 1∼4, an n-cube containing $2n-6$ or less conditional node-faults can be partitioned along a certain dimension so that both $(n-1)$-dimensional subcubes are conditionally faulty. As a consequence, we can apply an inductive proof to improve Fu's result [3] by showing that an conditionally faulty n-cube, with $f \leq 2n-6$ faulty nodes, contains a fault-free path of length at least $2^n - 2f - 1$ (respectively, $2^n - 2f - 2$) between any two nodes of odd (respectively, even) distance. In fact, we believe that many fault-tolerance related issues on hypercubes can be concerned on the other hypercube-like interconnection networks, such as crossed cubes, twisted cubes, etc. Hence, it is also intriguing for us to do such an extension as future work.

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